***Grade 11 | Semester 1***

# CAPSTONE CHALLENGE

**2026–2027 | Grade 11 | Semester 1**

## Smart Food Packaging: Integrating Passive Preservation with Natural Quality Indicators

### Theme: Change, Equilibrium, and Cycles

### 1. Egypt Grand Challenges

- Recycle garbage and waste for economic and environmental purposes.
- Work to eradicate diseases and address public health issues.
- Address and reduce pollution fouling our air, water, and soil.
- Increase the industrial and agricultural bases of Egypt.

### 2. Capstone Big Idea

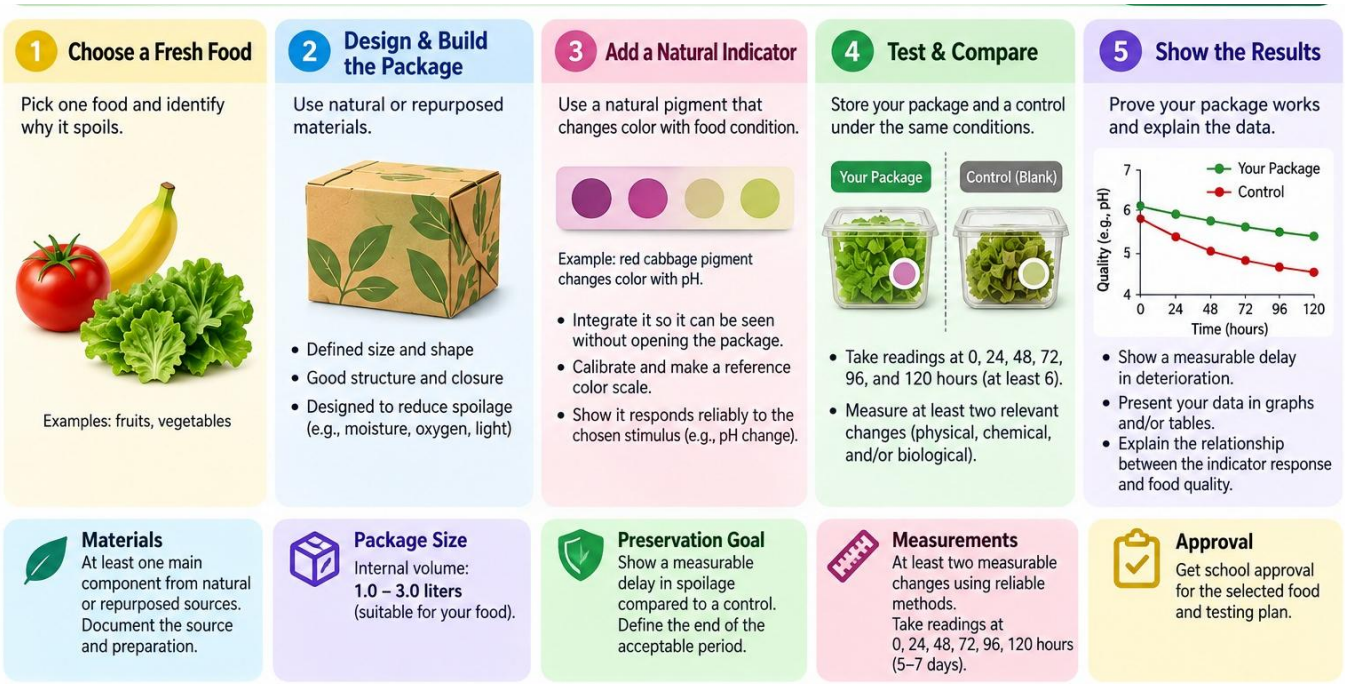
Food packaging can influence the rate of physical, chemical, and biological deterioration by controlling the environment surrounding food. At the same time, changes in fresh food or its surrounding environment can produce measurable signals. Students must harness natural and/or repurposed materials to develop useful eco-friendly packaging components and a natural sensing indicator, creating one integrated system that both preserves fresh food and communicates a measurable change in the condition of the food.

### 3. Essential Question

How can natural or repurposed materials be engineered into food packaging that delays physical, chemical, and biological deterioration and is equipped with a smart label that senses and communicates a measurable change in the condition of the food?

### 4. Design Challenge

Students must investigate why a selected fresh food deteriorates during storage and how packaging can slow the relevant deterioration mechanisms. Each team must design and fabricate one portable packaging prototype for fresh food, primarily from natural and/or repurposed materials. The prototype must have defined geometry, adequate structural integrity, an appropriate closure/sealing mechanism, and a justified preservation function. The prototype must also include a Product Validation Indicator (PVI) derived from a natural pigment extracted from an appropriate source. The indicator must respond to a recognizable physical, chemical, and/or biological change associated with a measurable change in the fresh food or package environment. Students must calibrate the indicator, conduct a controlled preservation experiment, compare the prototype with a control/blank, and demonstrate that the packaging provides a measurable preservation benefit. Students must demonstrate a scientifically justified relationship between the indicator response and selected measurements of fresh-food quality or condition. Scheme 1 shows the challenge workflow.




**Scheme 1.** Capstone challenge workflow.

## 5. Design Requirements

### 1. Food and application

Each team must select one fresh food product or category and identify its main deterioration mechanisms. The school must approve the selected food and testing procedures before experimentation.

### 2. Material requirement

At least one major functional component of the packaging system must be derived from a natural and/or repurposed eco-friendly material stream. Students must document the source, preparation, treatment/transformation, and reason for selecting the material. The prototype must also include a measurable sensing indicator based on a pigment material derived from natural and/or repurposed sources.

The indicator must be integrated so that the evaluator can observe it without opening or damaging the package. The team must establish a reference sensing scale. Before preservation testing, students must demonstrate that the sensing material responds reproducibly to controlled changes in the selected stimulus.

### 3. Packaging characteristics

The package must be designed to reduce at least one relevant driver of deterioration, such as moisture exchange, oxygen exposure, light exposure, or other scientifically justified factors. Students must report material composition/source, external/internal dimensions, internal volume, wall/layer thickness, total mass, closure/sealing method, structural configuration, and number/type of layers, and explain how these characteristics support preservation.

### 4. Maximum and minimum prototype dimensions

The prototype volume must range from 1.0 to 3.0 L. The intended internal volume must suit the selected fresh food and be justified by the team.

### 5. Preservation requirement

The package must demonstrate a measurable delay in deterioration relative to a control/blank condition. Before testing, the team must define what constitutes the end of the acceptable preservation period.

### 6. Requirements for measurable changes in food

Students must select at least two measurable changes (physical, chemical, and/or biological) that are scientifically relevant and feasible for the selected fresh food. Students must use the selected measurable changes and justified indicators with teacher approval.

### 7. Experimental measurement requirements

For each selected quantitative preservation indicator, collect **at least six readings** for each measurable change at defined storage times, including the initial condition: 0, 24, 48, 72, 96, and 120 h. **The recommended test period is 5–7 days**. If the predefined endpoint is reached earlier, document the endpoint and the reason for stopping. The measurements must be conducted using a calibrated device, a reference method, a standard, repeated measurements, or a scientifically justified comparison.

## 6. Constraints

- All material preparation and testing must be conducted in the school laboratory or other facilities approved by the school.
- The prototype must use safe, natural, and/or repurposed-derived materials and must not contain radioactive materials or hazardous chemicals.
- Students must not intentionally culture or amplify potentially pathogenic microorganisms.
- Food samples exposed during experiments must not be consumed after testing.
- Any material intended to contact food directly must be approved for food-contact use; otherwise, an approved food-contact barrier must separate the food from the repurposed material.
- The sensing indicator must not be placed in direct contact with food unless direct food contact has been demonstrated to be safe and approved by the school.
- Testing conditions must be documented and kept consistent between prototype and control samples.
- When a measurement is destructive, separate matched samples/packages must be prepared for the corresponding measurement time rather than repeatedly opening the same preservation package.
- All components required for the exhibition demonstration are the responsibility of the team.
- The project must not claim that the indicator independently certifies food safety; conclusions must be limited to the variables actually measured and validated.
- Any device used in data collection must be calibrated using a standard technique.

## 7. Evaluation-day demonstration

- The team must present a sealed prototype containing a representative, approved food sample with a measurable indicator. The evaluator must be able to interpret the indicator using the team's reference scale without opening the package. During the evaluation session, the team should present prepared test samples representing at least three previously characterized conditions (e.g., fresh/acceptable, intermediate, and deteriorated or simulated-deteriorated). Without opening the package, the evaluator should use the team's indicator/blank scale to identify the condition.
- Students must present time-dependent graphs showing changes in fresh-food quality and indicator responses. At least one graph must show the indicator response versus time or stimulus. Prototype and control/blank data should be shown together where appropriate.

The team must provide video evidence of material selection, preparation, packaging fabrication, PVI preparation, calibration, preservation testing, measurements, control comparison, data analysis, and final validation.